

Causally Verified QoS-Floor Constraints for Decentralized SAGIN Access Association

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Abstract—Decentralized access association in a three-tier space-air-ground network must protect a per-slot quality-of-service floor and not merely a long-run average, because a policy can meet the average while starving some vehicle in every slot. The standard remedy is to price that floor with a Lagrangian multiplier learned online, after which the price is taken to be working because it rises and settles. We show that this inference can be empty, and replace it with a causal test: cut the dual error signal, misdirect it across agents while preserving the multiplier trajectory pointwise, and reverse its sign—an effect that survives all three was never carried by the price. The test assumes nothing specific to this setting. A per-vehicle multiplier enforcing a fleet-relative floor appears to raise the per-slot floor $4.4\times$; the three controls expose that gain as an open-loop artifact of re-initializing, every episode, a price that provably cannot rise—the tell being a “constraint” that improved throughput and the floor at once, which a constraint cannot do. Budgeting one-sided expected shortfall instead restores a price that survives all three, raising the per-slot floor $17.1\times$ at $N=12$ and $63.1\times$ at $N=20$ over a matched unconstrained learner, cutting outage at 0.1 Mbps from 35.0% to 7.2%, and beating proportional-fair and greedy association on throughput, floor and outage at once.

Index Terms—space-air-ground integrated network, multi-agent reinforcement learning, access association, constrained MDP, Lagrangian methods, reproducibility, 6G

I. INTRODUCTION

Sixth-generation networks are expected to integrate low-Earth-orbit (LEO) satellites, high-altitude platform stations and terrestrial base stations into a single space-air-ground integrated network (SAGIN) [1]. No tier dominates for every vehicle at every instant, so vehicles associate dynamically, and when many share the spectrum the association couples: vehicles on the same tier interfere, so a purely greedy rule collapses the strongest layer. A centralized optimizer would need global channel state at vehicular timescales, so we consider *decentralized* policies learned by multi-agent reinforcement learning (MARL).

Driven by throughput alone, such policies starve the worst-served vehicles, and fairness here must be read at *two time scales*: over a long run even a random policy avoids starvation by rotating which vehicle suffers, but within any one slot some vehicle may still be driven to a near-zero rate. For delay-sensitive traffic [2] it is this per-slot floor that determines service. The natural remedy is to make the floor a constraint

and price it with a Lagrangian multiplier updated online by dual ascent. What is rarely checked is whether the resulting price acts as feedback at all: a multiplier that rises and settles is taken as evidence that the constraint is being enforced. This paper is about what happens when that inference is wrong, and about how one would know.

A. Adaptation that is not feedback

Applied directly, the recipe appears to work: a per-vehicle multiplier enforcing a fleet-relative floor raises the per-slot rate floor $4.4\times$ over an unconstrained learner at $N=12$. It also raises sum throughput by 13.9 Mbps, and that second number is the problem: a constraint can only shrink the feasible set, so a gain on both counts is not the constraint’s. Following it back, the multiplier turns out not to rise at all, leaving the per-episode re-initialization as the only event that ever lifts the price. The “price” is thus a fixed schedule, high early and low late, replayed identically whatever the vehicles experience: open loop, not feedback. Remove the reset, or start the multiplier at zero, and the gain disappears.

This is not a bug report: the same reasoning applies to any constraint defined relative to a group aggregate, and the diagnosis needed a tool, since a degenerate multiplier still produces a plausible curve. *Constraint satisfaction is not self-evidencing*. We therefore ask what evidence distinguishes a price being learned from a penalty schedule that happens to work, and answer by intervening on the loop itself.

B. Related work

Learning-based resource management in SAGIN typically relaxes fairness into reward shaping or a time-averaged index. Closest to us, Yang *et al.* [3] apply constraint-aware MADDPG to three-tier uplink scheduling, but their Jain-index constraint becomes a hand-tuned shaped reward and their agents are network-side schedulers over orthogonal sub-channels rather than user-side agents under interference-coupled SINR; Wan *et al.* [4] treat per-UAV QoS in isolation. Pricing has also been used for coordination rather than constraint satisfaction: [5] broadcasts slow-timescale congestion prices, addressing contention rather than a per-vehicle floor.

Our mechanism is an instance of Lagrangian primal–dual safe RL, in which a constraint is relaxed into the reward

through a multiplier raised by dual ascent [6], and we adopt its established components rather than propose them: that an expectation constraint does not imply per-step satisfaction, and that one-sided or risk-based costs are the remedy, is standard [7], [8]; that the plain dual is sensitive to initialization and step size is due to [9]; that a converged primal–dual solution coincides with a fixed-penalty optimum is [10]. What none covers is the question we ask: *how does one check that a learned multiplier is functioning as feedback at all?* This is the constrained-RL analog of a documented pattern in deep RL, where reported gains were traced to implementation details rather than the mechanism claimed [11]; we know of no such check for constrained multi-agent RL.

C. Contributions

- **A constraint whose price provably cannot rise.** Whenever a floor is defined relative to a group aggregate and relaxed by a *signed* residual, that residual aggregates to $(\alpha - 1)\bar{r} \leq 0$ identically. This is algebra, not an implementation defect: it holds for any such constraint, and per-vehicle multipliers do not escape it, because violations rotate and leave every λ_k on the projection boundary at once. The $4.4\times$ gain such a mechanism reports survives neither removing the per-episode reset nor starting the multiplier at zero (§III-B, §IV-B).
- **A causal protocol for deciding whether a multiplier is feedback.** We corrupt the loop three ways: the error signal *cut*, *misdirected* across agents, or *reversed* in sign, and ask whether the effect survives. The reversed arm is the sharpest: its multiplier is adaptive by every measurable property and aimed at the right vehicle, yet the floor still collapses, so what the loop needs is not that the price moves but that its movement carries the sign of the constraint error. *The same controls return opposite verdicts on the two formulations* (§IV-C).
- **A constraint that restores an informative signal, and what it costs.** We budget one-sided expected shortfall, an established remedy for average constraints that miss the tail [7], [8]; §III-B shows one is *required* here rather than merely preferable. The budget is fixed before training from an untuned reference, and the operating point tracks it across five budgets, from infeasible through tight to slack. At equal throughput it lands *on* the tuned fixed-penalty frontier, which we report as a boundary rather than a shortfall (§IV-D).
- **The wireless result.** The repaired policy raises the per-slot floor $17.1\times$ at $N=12$ and $63.1\times$ at $N=20$ over a matched unconstrained learner and cuts outage from 35.0% to 7.2% for 12.7% of sum throughput, beating proportional-fair and greedy association on all three at once (§IV-E).

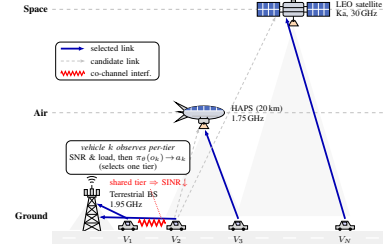


Fig. 1. Vehicle-side access association in a three-layer SAGIN. Vehicles sharing a tier (here V_1, V_2 on the BS) interfere on the uplink (red); shaded cones sketch the coverage-capacity trade-off.

II. SYSTEM MODEL

A. Network Architecture

We consider a three-layer SAGIN serving N ground vehicles, each acting as a decision-making agent (Fig. 1). We model *one node per tier*, so association is a choice among tiers rather than among nodes. Each uplink tier sits in a band that regulation actually assigns to it in this direction, so cross-tier co-channel interference is excluded by construction rather than by assumption: a LEO satellite at 600 km on the Ka-band uplink (30 GHz, NTN reference Set-1 [12]), a high-altitude platform station (HAPS) at 20 km on 1.75 GHz (the range ITU RR footnote 5.388A assigns to HAPS *reception*), and a terrestrial base station on 1.95 GHz (3GPP Band 1 uplink). Each slot, vehicle k selects one tier $a_k^t \in \{\text{LEO}, \text{HAPS}, \text{BS}\}$ for its *uplink*; we study vehicle-side association over access links, not feeder or backhaul selection, in the dense hot-spot regime where all three tiers overlap.

B. Channel and Interference Model

Elevation is explicit because it decides which tier is “capacity” and which “coverage”: the LEO tier sits 20.2 dB above the BS tier at 90° and 23.3 dB below it at 10° . We report at 50° and re-test across elevation the one claim that depends on it (§IV-D). The two non-terrestrial tiers use free-space path loss with the urban LoS-probability and clutter-loss tables of [13] at the nearest reference elevation, each vehicle drawing its own blockage state; the ground tier uses UMa [14]. Antenna gains and the remaining parameters are given in §IV; the vehicle takes the VSAT class on the satellite link, the “moving platforms” column of [12] being unspecified.

Vehicles on the same tier share its spectral resource and interfere on the uplink. With the association indicator and per-tier load

$$x_{k,a}^t = \mathbf{1}[a_k^t = a], \quad L_a^t = \sum_{k=1}^N x_{k,a}^t, \quad (1)$$

the uplink SINR and achievable rate of vehicle k are

$$\text{SINR}_{k,a}^t = \frac{P g_{k,a}^t}{\sum_{j \neq k} x_{j,a}^t P g_{j,a}^t + \sigma^2}, \quad (2)$$

$$r_{k,t} = \sum_{a \in \mathcal{A}} x_{k,a}^t \log_2(1 + \text{SINR}_{k,a}^t), \quad (3)$$

where $g_{k,a}^t$ absorbs path loss, blockage, fading and antenna gains, all vehicles use equal normalized power P , and σ^2 is the noise power. The constrained quantity is the spectral efficiency $r_{k,t}$, which is what the floor and the shortfall budget of §III act on; the delivered rate is $B r_{k,t}$ with $B = 20$ MHz identical across tiers, and throughput, floors and outage are reported in Mbps. The load L_a^t enters the denominator of every vehicle on tier a , the coupling that defeats purely greedy selection.

III. PRICED SHORT-TERM QOS FOR DECENTRALIZED ACCESS ASSOCIATION

A. Constrained Markov game

Each vehicle k selects a tier every slot from its own observation and receives the spectral efficiency $r_{k,t}$ of (3). Let $\tilde{r}_{k,t}$ be an exponentially weighted moving average of $r_{k,t}$, and let

$$\bar{x}_t = \alpha \cdot \frac{1}{N} \sum_j \tilde{r}_{j,t}, \quad \alpha \in (0, 1), \quad (4)$$

be a floor defined *relative* to the fleet. The objective is to maximize mean throughput while keeping every vehicle near $\bar{x}_t$ in *individual slots*. A relative floor keeps one α meaningful as load and geometry change the achievable rate by orders of magnitude. We report Jain's index [15] but do not constrain it: being scale-invariant, it calls a uniformly starved allocation "fair".

B. What a signed constraint cannot see

The obvious relaxation constrains the rate itself, $g_k = \mathbb{E}[\bar{x}_t - \tilde{r}_{k,t}] \leq 0$, and prices it by projected dual ascent. That residual is *signed*: positive when a vehicle is below the floor, negative without bound when it is above. It is not the constraint the objective asks for, and it fails in two separable ways.

Proposition 1 (Signed aggregation cancels the residual). *For any $\alpha < 1$ and any rate profile,*

$$\frac{1}{N} \sum_k (\bar{x}_t - \tilde{r}_{k,t}) = (\alpha - 1) \cdot \frac{1}{N} \sum_j \tilde{r}_{j,t} \leq 0. \quad (5)$$

A single multiplier driven by this mean is therefore monotone non-increasing and reaches the projection boundary however badly individual vehicles are served; the cancellation is exact and has nothing to do with learning. The claim is narrow on purpose—it concerns *signed linear* aggregation, and replacing the sum in (5) by $\max_k$, by a mean of hinges, or by a conditional value-at-risk (CVaR) constraint removes the identity.

Per-vehicle multipliers escape Proposition 1, since an individual residual is positive whenever that vehicle is starved, but they fail for a second reason that is a modeling error rather than an algorithmic one. That an expectation constraint does not imply per-step satisfaction is well known in safe RL, and risk measures such as CVaR are the standard response [7], [8].

Proposition 2 (Signed expectations are blind to intermittent starvation). *There are rate processes for which every per-vehicle constraint $\mathbb{E}[\bar{x}_t - r_{k,t}] \leq 0$ is satisfied strictly, while in every slot some vehicle receives zero rate.*

Proof. Take $N = 2$ with $(r_{1,t}, r_{2,t}) = (0, 2)$ for odd t and $(2, 0)$ for even t . The fleet mean is 1 every slot, so with $\alpha = 0.25$ each residual is 0.25 half the time and -1.75 the other half, giving $\mathbb{E}[\bar{x}_t - r_{k,t}] = -0.75 < 0$ while each vehicle is starved in half of all slots. Smoothing does not repair this: since $\mathbb{E}[\tilde{r}_k] = \mathbb{E}[r_k]$ in steady state, the EWMA form takes the same value. $\square$

The constraint is thus not failing to be enforced; it is the wrong constraint, and unlike Proposition 1 this has nothing to do with there being several agents. What the fleet-relative reference adds is concealment: violations rotate, so every vehicle's time-averaged residual is negative *simultaneously* and all λ_k sit on the projection boundary at once, reporting feasibility together. At $N=12$, $\alpha=0.25$ a uniform-random reference over five seeds is in violation in only 0.6–5.3% of slots per vehicle, yet all 60 time-averaged residuals are negative, in $[-0.731, -0.528]$: no learning is needed. Sweeping α up to 0.50 leaves the converged multipliers below 0.005 (training-log protocol).

C. Budgeting expected shortfall

We instead constrain the *shortfall* the reward already penalizes,

$$\mathbb{E}_\pi[\underbrace{(\bar{x}_t - r_{k,t})_+}_{c_{k,t}}] \leq \epsilon \mathbb{E}_\pi[\bar{x}_t], \quad k = 1, \dots, N, \quad (6)$$

with the instantaneous rate inside the hinge, since that is the per-slot event the long-term index hides. One-sided costs are standard [7], [8]; §III-B supplies the reason one is *required* here. Being non-negative, $c_{k,t}$ escapes Propositions 1 and 2, so the equilibrium is set by the budget ϵ —dimensionless, and readable as *a vehicle may fall short of the floor by at most a fraction ϵ of it, on average*—rather than by the sign of the signal. Pricing the rate residual while penalizing the hinge would relax two different constraints, since $\mathbb{E}[(a - r)_+] \geq a - \mathbb{E}[r]$, strictly whenever the residual changes sign.

Because the right-hand side of (6) is itself policy-dependent, the Lagrangian carries it too. With μ weighting handovers, vehicle k 's reward and price update are

$$R_{k,t} = \frac{1}{N} \sum_j r_{j,t} - \lambda_k (c_{k,t} - \epsilon \bar{x}_t) - \mu \mathbf{1}[a_{k,t} \neq a_{k,t-1}], \quad (7)$$

$$\lambda_k \leftarrow [\lambda_k + \eta (c_{k,t} - \epsilon \bar{x}_t)]_+. \quad (8)$$

The bracketed quantity is the same in both, so the policy is charged exactly the constraint the price tracks; omitting $\epsilon \bar{x}_t$ from (7) would discard a real gradient, since a policy can relax its own constraint by lifting the fleet mean. Multipliers persist across episodes.

D. A budget is a specification; a penalty weight is not

Why is (8) needed when a fixed penalty with a hand-chosen λ is simpler? Not because dual ascent reaches a better trade-off—at convergence it should not, and §IV-D confirms it does not—but because a penalty coefficient cannot see the service

level it is meant to deliver: for a *constant* budget c it optimizes $R(\pi) - \lambda C(\pi) + \lambda c$, whose last term is independent of π . That algebra does not carry over to (6) unchanged, since our right-hand side is policy-dependent, so whether the budget is tracked by the dual or by the reward term is an empirical question that §IV-C settles by freezing the multiplier. What the dual supplies, if anything, is the map $c \mapsto \lambda^*(c)$, which an operator stating a service level does not have.

E. Fixing the budget without a sweep

This still leaves ϵ to choose, so we fix it before training from a policy with nothing to tune. Let

$$\epsilon = \frac{\mathbb{E}_{\pi_{\text{greedy}}} \left[\frac{1}{N} \sum_k (\bar{x}_t - r_{k,t})_+ \right]}{\mathbb{E}_{\pi_{\text{greedy}}} [\bar{x}_t]}, \quad (9)$$

the normalized mean shortfall of greedy-SNR association, measured on the environment alone. The constraint then reads: *the priced policy may not fall short of the floor more, on average, than an untuned baseline does.* Over load this is nearly flat—0.336, 0.361, 0.377, 0.383, 0.379 at $N = 6, 8, 12, 16, 20$ —which makes (9) a rule rather than a per-load fit, and barely depends on which untuned reference is used (0.393 for proportional fair at $N=12$ against greedy’s 0.387). Two aggregations of (9) differ by 2.6% (0.377 averaging per-slot ratios, 0.387 as the ratio of expectations that (8) equilibrates). We budget the smaller and report the larger, so every achieved-to-target ratio below is pessimistic by that margin.

F. Learning

The priced reward (7) is optimized with MAPPO [16] under centralized training and decentralized execution; observations are per-vehicle solo SNRs, per-tier occupancy shares and the previous action; the shared actor takes a one-hot identity input, so adding a vehicle adds no parameters. (8) runs environment-side at the slot timescale: the price shapes training without entering the observation or the deployed policy.

IV. EXPERIMENTS

A. Setup and protocol

Vehicles number $N \in \{6, 8, 12, 16, 20\}$ over 200-slot episodes. Uplink carriers are 30, 1.75 and 1.95 GHz for the LEO, HAPS and BS tiers at 600 km, 20 km and ground level, at 50° elevation, with 20 MHz of bandwidth, −100 dBm noise and 81.7/16.5/8.0 dBi of combined antenna gain; path loss is FSPL plus NTN LoS/clutter [13] on the non-terrestrial tiers and UMa [14] on the ground tier, with Nakagami- m ($m=1$) fading and log-normal shadowing. The constraint uses $\alpha = 0.25$, EWMA $\beta = 0.9$, handover weight $\mu = 0.05$, dual step $\eta = 5 \times 10^{-3}$, $\lambda_{\text{init}} = 1$, $\lambda_{\text{max}} = 30$, and ϵ from (9) at each load, untuned. The actor is a GRU(128) with a one-hot identity input, the critic is central, and PPO runs at learning rate 3×10^{-4} , clip 0.2, $\gamma = 0.99$, over 10 environments for 10^5 steps. Every number below comes from one protocol: 500 evaluation episodes of 200 slots on held-out seeds. The one

TABLE I
COMPARISON AT $N=12$ (EVAL; MBPS; OUTAGE AT 0.1 MBPS; JFI IS JAIN’S FAIRNESS INDEX; SWITCH IS THE PROBABILITY A VEHICLE CHANGES TIER BETWEEN CONSECUTIVE SLOTS). ROWS ARE INDEPENDENT MEASUREMENTS OVER ALL SEEDS, SO DIVIDING TWO IS NOT THE MATCHED-SEED COMPARISON WE CLAIM: SLOT-MIN GIVES $17.4 \times$ AGAINST THE MATCHED-SEED TEXT VALUE $17.1 \times$. PID-LAGRANGIAN ($k_p=1$) RUNS OUR BUDGET AND IS FLOOR-EQUIVALENT, SO NO CELL IS MARKED BEST.

Method	Sum TP	Slot-min	P_{out}	JFI	Switch
Proposed ($\epsilon=0.377$)	222.7	0.1917	0.072	0.254	0.518
PID-Lagrangian [9]	221.0	0.2007	0.070	0.257	0.528
Unconstrained MAPPO	252.2	0.0110	0.371	0.190	0.312
Direct form (artifact)	266.1	0.0482	0.157	0.208	0.355
Jain-shaped [3]	248.9	0.0109	0.389	0.192	0.327
Fixed penalty $\lambda=1$	228.1	0.1541	0.085	0.251	0.480
Greedy-SNR	206.7	0.1579	0.091	0.255	0.585
Proportional fair [17]	213.6	0.1562	0.083	0.248	0.544

exception is the converged multiplier, which does not exist at evaluation time and is read from training logs, labeled as such. Comparisons are paired by seed and tested with an exact sign-permutation test. With n seeds the smallest attainable two-sided p is 2^{1-n} : $p=0.0625$ at $n=5$ and $p=0.0078$ at $n=8$ are floors, not weak evidence. Claims that two variants behave the *same* use two one-sided tests (TOST), not a failure to reject. The margin is fixed in advance at one tenth of the constrained-versus-unconstrained effect, computed from two arms taking part in no comparison.

B. The reported gain is an artifact

The direct formulation of §III-B reports a per-slot floor $4.37 \times$ that of a matched unconstrained learner at $N=12$ and $3.49 \times$ at $N=20$, together with +13.9 and +22.8 Mbps of sum throughput: a mechanism improving the floor and throughput together is not acting as a constraint. Because the multiplier provably cannot rise (Proposition 1, and 1592 consecutive within-episode steps with zero increases; Fig. 2), the per-episode re-initialization is the only event that ever lifts it, making the penalty a high-early, low-late schedule repeated every episode. A 2×2 ablation confirms that removing either ingredient removes the entire gain: at $N=12$, starting the multiplier at zero gives $1.00 \times$ the unconstrained floor, carrying it across episodes $0.97 \times$, and doing both $0.99 \times$ (sum throughput +0.6, −1.8 and −0.3 Mbps); at $N=20$ the same three arms give 0.99 , 0.84 and $1.00 \times$.

C. Testing whether the multiplier is feedback

A curve that rises and settles is not evidence that a price is being learned. The interventions below are not arbitrary: for a multiplier to act as feedback, three conditions must hold together: the constraint error must reach it, it must be aimed at the vehicle that incurred that error, and it must move in the direction the error implies. Each intervention removes exactly one of the three and leaves the others intact, which is what makes Table II a decision rule rather than an ablation: an effect that survives all three was never carried by the price.

Cut freezes the multiplier at the value an oracle sweep selects for the anchor budget and sweeps only the budget. The

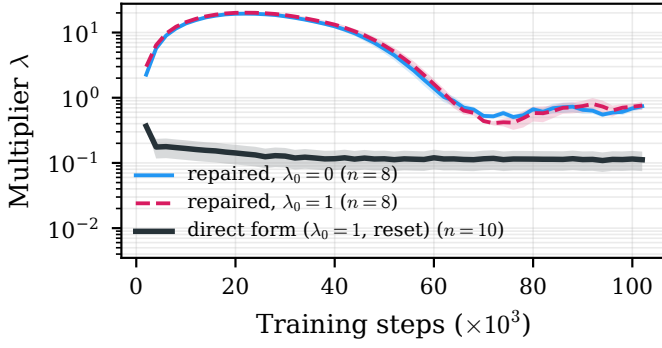


Fig. 2. Multiplier trajectories at $N=12$, mean over seeds with ± 1 s.e. bands (training-log protocol: the one quantity that cannot be reported under the EVAL protocol used everywhere else). The direct formulation, run as reported earlier ($\lambda_0=1$ with a per-episode reset), never rises above the value it starts each episode from, as Proposition 1 requires. The repaired formulation rises while the budget is violated and settles once it is met; its two curves lie on top of one another, which is the point—the operating point does not depend on where λ starts. Dual step sizes are those of each formulation as run, 0.005 for the direct form and 0.5 for the repaired one.

TABLE II

THE THREE INTERVENTIONS READ AS A DECISION RULE ($N=12$, EVAL): AN EFFECT SURVIVING AN INTERVENTION WAS NOT RELYING ON THE PROPERTY THAT INTERVENTION REMOVES. ENTRIES ARE THE PER-SLOT FLOOR AS A MULTIPLE OF THE UNCONSTRAINED LEARNER OVER THE SEEDS EACH COLUMN SHARES, SO THE DEGENERATE INTACT LOOP READS $4.51 \times$ HERE AGAINST $4.37 \times$ IN §IV-B, WHICH PAIRS OVER A DIFFERENT SET. [†]FREEZING λ REMOVES BUDGET-FOLLOWING RATHER THAN THE FLOOR, SO THAT ARM IS SCORED BY THE SLOPE OF ACHIEVED ON TARGET SHORTFALL (1 PERFECT, 0 NONE). DASHES MARK ARMS NOT RUN.

Arm	Breaks	Preserves	Degen.	Repaired
Intact loop	—	—	4.51	17.06
Cut (frozen λ)	error signal	—	—	0.07 [†]
Misdirect (yoked)	targeting	trajectory	4.41	0.99
Reverse (anti)	sign	signal, aim, step	—	1.07
Unconstrained	—	—	1.00	1.00

regression slope of achieved on target shortfall is 0.07, against 0.92 for the adaptive dual over the same span: adaptation supplies $\approx 92\%$ of the tracking, the policy-dependent right-hand side of §III-D the rest. Against a static controller alone this would state a definition, not a finding; the next two answer that. **Misdirect** preserves the multiplier trajectory pointwise and permutes the residual across vehicles, so the policy pays the same prices at the same times, aimed at the wrong vehicle. **Reverse** flips the sign of the dual step alone, leaving signal, targeting and step size intact; that multiplier stays adaptive and correctly aimed, and the floor collapses anyway.

The two formulations separate under the same controls (Table II): on the repaired mechanism every intervention returns the unconstrained baseline while the intact loop does not, whereas on the degenerate one misdirection reproduces 98% of the reported gain, that mechanism never having used vehicle-specific feedback at all.

Both collapses reach the floor of the exact sign-permutation test (-0.1806 and -0.1796 against the intact loop, $p = 0.0078$ at $n=8$).

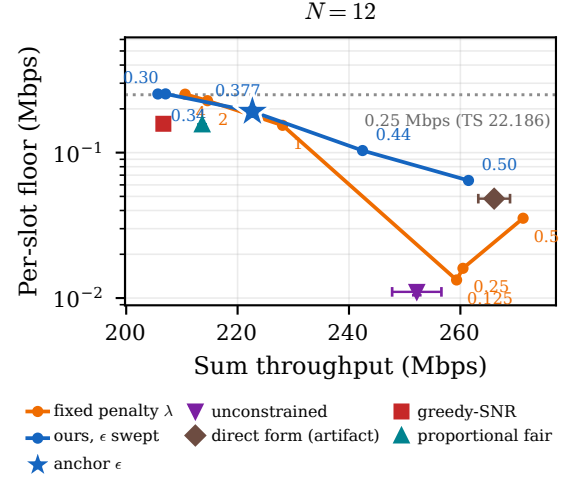


Fig. 3. Throughput against per-slot floor at $N=12$ (EVAL, arm means as in Table I). Both curves are swept knobs— ϵ for ours, λ for the fixed penalty—and they coincide. The asymmetry is that the star marks the ϵ that (9) selects before training, from a baseline measurement and without reference to this plane; no comparable rule selects a λ .

D. What the restored constraint delivers, and what it costs

With the shortfall budget in place the dual converges to an interior value independent of whether it starts from 0 or 1 (Fig. 2; TOST $p = 0.006$, $n=8$, at the pre-declared margin); a tenfold change of the dual step size moves the converged price by 1.7%, from 0.5304 to 0.5212 at the anchor budget (training-log protocol).

The operating point follows the budget it is given. Over $\epsilon \in \{0.30, 0.34, 0.377, 0.44, 0.50\}$ realized shortfall stays within 10% of target (achieved/target 0.94–1.08) and the three KKT regimes appear in turn: at 0.30 the constraint is infeasible and the multiplier rests on its ceiling; between 0.34 and 0.44 it is tight and the multiplier interior; at 0.50 it is slack and the multiplier small. Behavior consistent with complementary slackness therefore holds across budgets, not at one tuned point, bounded below by feasibility: realized shortfall never falls below ≈ 0.32 .

These measurements also bound what a learned price can deliver. At equal throughput the priced policy lands on the tuned fixed-penalty frontier, not above it ($1.00, 1.04, 1.07 \times$ at $N = 6, 12, 20$; Fig. 3), which is what a converged primal–dual solution should do [10]. Nor does the budget transfer better than a weight: one $\lambda \approx 2$ lands within 4% of our throughput and 20% of our floor at every load tested ($N = 6, 12, 20$), and the anchor rule is not itself stable across geometry: on the greedy reference its ϵ ranges over 0.41–0.61 between 10° and 70° of elevation. We make no transfer claim. What the dual changes is not the attainable trade-off but how one asks for a point on it: an operator states a tolerable shortfall and (8) finds the price, whereas a penalty coefficient must be searched for and means nothing at the service level. It is the constraint that matters, not the controller: a PID-Lagrangian multiplier [9] (Table I) and one shared price in place of N are both floor-

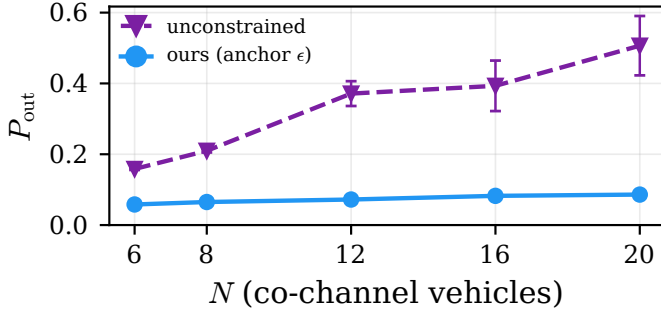


Fig. 4. Outage at 0.1 Mbps against load, each N at its own anchor budget from (9) and untuned; seed means, ± 1 s.e., EVAL protocol. The budget flattens the curve rather than shifting it.

equivalent at the declared margin (TOST $p \leq 0.012$), and the direct form fails under each.

E. Access performance

Against a matched unconstrained learner the priced policy raises the per-slot floor from $3.0\times$ at $N=6$ to $63.1\times$ at $N=20$ and cuts outage at 0.1 Mbps by $2.7\text{--}5.9\times$, at a throughput cost of 2.6% to 14.9% (all ratios on matched seeds): the gain grows with congestion, the regime a floor exists for. At $N=12$ it beats proportional fair and greedy-SNR on throughput, floor and outage at once (Table I), while unconstrained MAPPO buys its extra throughput with a floor two orders of magnitude lower, and Jain-index shaping is indistinguishable from it: a long-horizon index does not protect a slot. What the ratios hide is in Fig. 4: under the budget outage is near load-independent, 0.058 to 0.086 while N more than triples, where the unconstrained learner’s climbs from 0.158 to 0.507. That stability is bought with re-association: the unconstrained learner switches less (0.31 against our 0.52), and the switch rate rises from 0.41 of vehicle-slots at $\epsilon=0.50$ to 0.63 at 0.30. A three-slot minimum dwell imposed at evaluation removes half the switching and three quarters of the floor, for a tuned fixed penalty and for greedy-SNR as much as for us; greedy-SNR is stateless and never trained, so that drop is a property of the timescale, not a train–test mismatch. We claim no lower bound (no arm was trained under a dwell constraint), and these are selection changes, 2.6 per vehicle-second, not handover procedures: no signaling or interruption is modeled.

V. CONCLUSION

A learned multiplier that rises, settles and reports feasibility is not by itself evidence that a constraint is being enforced. In a three-tier SAGIN uplink a $4.4\times$ floor gain proved to be an artifact of re-initializing a multiplier that provably could not rise, and the tell was a constraint that appeared to increase throughput. Because a degenerate price still looks like one, we decide the question causally: cut the error signal, misdirect it, reverse its sign. The three controls return opposite verdicts on the broken and the repaired formulation without the answer being known in advance. Budgeting one-sided expected short-fall restores a price that survives all three and delivers a $17.1\times$

per-slot floor at $N=12$. It does not beat a well-tuned penalty weight at equal throughput. What it supplies is the map from a stated service level to the price delivering it, and nothing in the protocol is specific to access association. The study is bounded by *one node per tier* (no intra-tier selection or handover), orthogonal tiers and an uplink-only budget. Proposition 1 is algebraic and does not depend on how many nodes a tier contains. The measured gains do: further nodes would dilute the co-tier interference that makes the floor hard to hold, so the numbers here are the harder case.

ACKNOWLEDGMENT

This work was supported in part by the Japan Science and Technology Agency via the JST ASPIRE Grant (No. JPMJAP2326), and in part by the Waseda University Grant for Special Research Projects (No. 2026C-512).

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